

A Finite Addressed Realization of the Standard-Model Gauge Representation on $V_7 \otimes \mathbb{C}[\mathbb{Z}_3 \times \text{Dic}_6]$

UCD Mathematical Physics Program

September 2026

Abstract

Let

$$\mathcal{H}_{504} = V_7 \otimes \mathbb{C}[\mathbb{Z}_3] \otimes \mathbb{C}[\text{Dic}_6], \quad \dim \mathcal{H}_{504} = 504,$$

where V_7 consists of one center state and a six-site ring, and

$$\text{Dic}_6 = \langle a, x \mid a^{12} = 1, x^2 = a^6, xax^{-1} = a^{-1} \rangle.$$

We construct commuting color, weak-isospin and hypercharge actions on this finite addressed space. The six-leaf sector factors by the Chinese remainder theorem as $\mathbb{C}[\mathbb{Z}_6] \simeq \mathbb{C}^3 \otimes \mathbb{C}^2$, yielding an $\mathfrak{su}(3)$ color action that annihilates the center. The central subgroup $\langle a^6 \rangle \simeq \mathbb{Z}_2$ yields a two-state weak fiber, while the one-dimensional character $\chi(a) = 1, \chi(x) = -1$ supplies a chirality involution. With A denoting center/leaf parity, $B = L(a^6)$, and P_L the chirality projector, the electric charge, weak generator and hypercharge are

$$Q = \frac{B}{2} + \frac{A}{3} - \frac{1}{6}I, \quad T_3 = \frac{1}{2}P_L B, \quad Y = 2(Q - T_3) = \frac{2A - I}{3} + P_R B.$$

After suppressing address multiplicities, the resulting isotypic sectors are exactly

$$Q_L : (\mathbf{3}, \mathbf{2})_{1/3}, \quad L_L : (\mathbf{1}, \mathbf{2})_{-1}, \quad \nu_R : (\mathbf{1}, \mathbf{1})_0, \quad e_R : (\mathbf{1}, \mathbf{1})_{-2},$$

with the quark singlets $u_R : (\mathbf{3}, \mathbf{1})_{4/3}$ and $d_R : (\mathbf{3}, \mathbf{1})_{-2/3}$, and with the independent $\mathbb{Z}_3$ factor providing three family copies. We verify the complete gauge Lie algebra, the Standard-Model electric charges and hypercharges including a right-handed neutrino, all one-generation gauge and mixed anomaly cancellations, the boundary trace ratio $G_Y/G_2 = 5/3$, and the one-loop coefficients $b_3 = -7, b_2 = -19/6, b_Y = 41/6$.

1 Finite addressed space

Let $\{|c\rangle, |1\rangle, \dots, |6\rangle\}$ be an orthonormal basis of V_7 . Define

$$P_c = |c\rangle\langle c|, \quad P_\ell = I_7 - P_c, \quad A = P_\ell - P_c = I_7 - 2P_c.$$

Thus $A^2 = I_7$ and the eigenvalue $A = -1$ labels the center while $A = +1$ labels the leaf sector.

On $\mathbb{C}[\text{Dic}_6]$ let $L(g)$ be the left regular representation and set

$$B = L(a^6).$$

Because a^6 is central of order two,

$$B^2 = I_{24}.$$

The lifted operators A and B commute on $\mathcal{H}_{504}$.

Lemma 1.1 (Charge polynomial reduction). *Let $C = 2A + B$. Then*

$$\frac{C^3 - 4C - 3I}{18} = \frac{B}{2} + \frac{A}{3} - \frac{1}{6}I.$$

Proof. Since $A^2 = B^2 = I$ and $[A, B] = 0$,

$$C^3 = (2A + B)^3 = 8A + 12B + 6A + B = 14A + 13B.$$

Hence $C^3 - 4C - 3I = 6A + 9B - 3I$, which gives the result. $\square$

We therefore define

$$Q := \frac{B}{2} + \frac{A}{3} - \frac{1}{6}I. \quad (1)$$

Its four (A, B) eigenvalue pairs give

$$(-1, -1) \mapsto -1, \quad (-1, +1) \mapsto 0, \quad (+1, -1) \mapsto -\frac{1}{3}, \quad (+1, +1) \mapsto \frac{2}{3}.$$

2 Color from the six-leaf ring

The leaf space is the regular space of $\mathbb{Z}_6$. The Chinese remainder map

$$\mathbb{Z}_6 \longrightarrow \mathbb{Z}_3 \times \mathbb{Z}_2, \quad n \longmapsto (n \bmod 3, n \bmod 2)$$

is a group isomorphism. Consequently

$$P_\ell V_7 \cong \mathbb{C}_{\text{col}}^3 \otimes \mathbb{C}_{\text{ori}}^2.$$

If S_6 is the unit shift on the ring, then in the CRT basis

$$S_6 = X_3 \otimes X_2.$$

Let $t_a = \lambda_a/2$, $a = 1, \dots, 8$, be the Gell-Mann generators on $\mathbb{C}^3$, normalized by

$$\text{tr}(t_a t_b) = \frac{1}{2} \delta_{ab}.$$

Define their action on V_7 by

$$T_a^{(3)} = 0 \oplus (t_a \otimes I_2).$$

Theorem 2.1 (Native color algebra). *The operators $T_a^{(3)}$ satisfy the $\mathfrak{su}(3)$ commutation relations, annihilate the center, and obey*

$$\sum_{a=1}^8 \left(T_a^{(3)} \right)^2 = \frac{4}{3} P_\ell.$$

Thus the center is a color singlet and the leaf sector is a color triplet with a two-dimensional address multiplicity.

Proof. The result follows from the fundamental $\mathfrak{su}(3)$ representation on the first CRT factor and the identity action on the second. The quadratic Casimir of the fundamental representation is $C_F = 4/3$. $\square$

3 Weak fiber and chirality

Let $Z = \langle a^6 \rangle = \{1, a^6\} \simeq \mathbb{Z}_2$. Choose a transversal T for Dic_6/Z . Every group element is uniquely tz^b , $t \in T$, $b \in \{0, 1\}$, and hence

$$\mathbb{C}[\text{Dic}_6] \cong \mathbb{C}^{12} \otimes \mathbb{C}_Z^2.$$

In this factorization $B = L(a^6) = I_{12} \otimes \sigma_x$. Let $Z_B = I_{12} \otimes \sigma_z$ and define

$$\tau_1 = Z_B, \quad \tau_2 = -iBZ_B, \quad \tau_3 = B.$$

Then

$$[\tau_i, \tau_j] = 2i\epsilon_{ijk}\tau_k, \quad \tau_i^2 = I.$$

Different choices of transversal are related by a unitary change of the Z -fiber frame, so the resulting $\mathfrak{su}(2)$ representation is unitarily equivalent.

Define the dicyclic reflection character

$$\chi(a^k x^e) = (-1)^e,$$

and the corresponding diagonal involution

$$\Gamma_\chi |a^k x^e\rangle = (-1)^e |a^k x^e\rangle.$$

Since $\chi(a^6) = 1$, Γ_χ commutes with the central Z fiber. For an orientation label $\sigma = \pm 1$, set

$$P_L = \frac{I - \sigma\Gamma_\chi}{2}, \quad P_R = I - P_L.$$

Finally define

$$T_i = \frac{1}{2}P_L\tau_i. \tag{2}$$

Theorem 3.1 (Chiral weak representation). *The operators in (2) satisfy*

$$[T_i, T_j] = i\epsilon_{ijk}T_k,$$

and vanish on $P_R\mathcal{H}_{504}$. Thus $P_L\mathcal{H}_{504}$ carries weak doublets while $P_R\mathcal{H}_{504}$ carries weak singlets.

Proof. P_L commutes with each τ_i and is a projector. Hence

$$[T_i, T_j] = \frac{1}{4}P_L[\tau_i, \tau_j] = i\epsilon_{ijk}\frac{1}{2}P_L\tau_k = i\epsilon_{ijk}T_k.$$

Also $P_RT_i = 0$. □

4 Hypercharge and particle representation

Set $T_3 = P_L B/2$ and define Y by the Gell–Mann–Nishijima relation

$$Q = T_3 + \frac{Y}{2}.$$

Using (1),

$$Y = 2(Q - T_3) = \frac{2A - I}{3} + P_RB. \tag{3}$$

Theorem 4.1 (Gauge quantum numbers). *After suppressing address multiplicities, the simultaneous (A, B, P_L) sectors carry the following representations of $SU(3)_c \times SU(2)_L \times U(1)_Y$:*

sector	A	B	$SU(3)_c$	$SU(2)_L$	Y
$L_L = (\nu_L, e_L)$	-1	± 1	1	2	-1
e_R	-1	-1	1	1	-2
ν_R	-1	+1	1	1	0
$Q_L = (u_L, d_L)$	+1	± 1	3	2	1/3
d_R	+1	-1	3	1	-2/3
u_R	+1	+1	3	1	4/3

The electric charges are respectively

$$Q(\nu) = 0, \quad Q(e) = -1, \quad Q(u) = \frac{2}{3}, \quad Q(d) = -\frac{1}{3}.$$

The independent factor $\mathbb{C}[\mathbb{Z}_3]$ supplies three family copies of this representation.

Proof. On P_L , equation (3) becomes

$$Y_L = \frac{2A - I}{3},$$

which is -1 for the center and $1/3$ for the leaves, independent of B as required for a weak doublet. On P_R ,

$$Y_R = \frac{2A - I}{3} + B.$$

Substitution of $A = \pm 1$ and $B = \pm 1$ gives the four singlet hypercharges shown in the table. The color statement follows from the previous theorem. $\square$

5 Address multiplicities

The finite addressed representation contains additional factors on which the gauge representation acts trivially. In particular, after fixing family, chirality and weak component, the lepton sectors have a six-dimensional residual address factor, while the quark sectors have

$$\mathbb{C}_{\text{col}}^3 \otimes \mathbb{C}_{\text{ori}}^2 \otimes \mathbb{C}_{\text{addr}}^6.$$

The gauge representation type is obtained by suppressing these spectator address factors. This distinction separates finite provenance multiplicity from particle-species multiplicity.

6 Anomaly cancellation

Use the left-handed Weyl convention. One family, including a right-handed neutrino, is

$$\begin{aligned} Q_L &: (\mathbf{3}, \mathbf{2})_{1/3}, & L_L &: (\mathbf{1}, \mathbf{2})_{-1}, \\ \nu_R^c &: (\mathbf{1}, \mathbf{1})_0, & e_R^c &: (\mathbf{1}, \mathbf{1})_2, \\ u_R^c &: (\bar{\mathbf{3}}, \mathbf{1})_{-4/3}, & d_R^c &: (\bar{\mathbf{3}}, \mathbf{1})_{2/3}. \end{aligned}$$

The anomaly coefficients are

$$\mathcal{A}_{SU(3)^2 Y} = 0, \quad \mathcal{A}_{SU(2)^2 Y} = 0, \quad \mathcal{A}_{\text{grav}^2 Y} = 0, \quad \mathcal{A}_{Y^3} = 0.$$

There are four left-handed $SU(2)$ doublets per family after color multiplicity, so the global Witten anomaly is absent.

7 Trace metric and one-loop running

For one reduced family and generators normalized by $\text{tr}(t_a t_b) = \delta_{ab}/2$,

$$G_3 = \text{Tr}(T_c^2) = 2, \quad G_2 = \text{Tr}(T_w^2) = 2, \quad G_Y = \text{Tr}\left(\frac{Y}{2}\right)^2 = \frac{10}{3}.$$

Hence

$$G_3 : G_2 : G_Y = 1 : 1 : \frac{5}{3}.$$

For a common parent kinetic normalization this gives

$$g_3^2 = g_2^2, \quad \frac{g_Y^2}{g_2^2} = \frac{3}{5}, \quad \sin^2 \theta_W = \frac{3}{8}$$

at the normalization scale.

With three families and one complex Higgs doublet of $Y_H = 1$, the one-loop coefficients in

$$\frac{dg_i}{d \ln \mu} = \frac{b_i}{16\pi^2} g_i^3$$

are

$$\boxed{b_3 = -7, \quad b_2 = -\frac{19}{6}, \quad b_Y = \frac{41}{6}.$$

For $g_1 = \sqrt{5/3} g_Y$, $b_1 = 41/10$.

8 Computational verification

The finite matrices were constructed directly in dimension 504. The following residuals were obtained in double precision:

$$\begin{aligned} \|Q_{\text{polynomial}} - Q_{\text{linear}}\|_F &= 0, \quad \|Y - Y_{\text{closed}}\|_F = 0, \\ \max_{i,j} \|[T_i, T_j] - i\epsilon_{ijk} T_k\|_F &= 0, \\ \max_{a,i} \|[T_a^{(3)}, T_i]\|_F &= 0, \quad \max_a \|[Y, T_a^{(3)}]\|_F = 0, \quad \max_i \|[Y, T_i]\|_F = 0. \end{aligned}$$

The color Casimir residual is 3.14×10^{-16} . The complete machine-readable derivation and independent verifier accompany this manuscript.

9 Conclusion

The finite addressed Hilbert space $V_7 \otimes \mathbb{C}[\mathbb{Z}_3 \times \text{Dic}_6]$ supports an explicit commuting representation of $\mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$. The center/leaf involution determines lepton versus quark hypercharge, the central dicyclic involution supplies weak T_3 , the reflection character supplies chirality, the six-leaf CRT factor supplies color, and the independent $\mathbb{Z}_3$ factor supplies three families. The resulting reduced representation has the Standard-Model gauge quantum numbers with a right-handed neutrino and is anomaly free.

References

- [1] J. Kogut and L. Susskind, Hamiltonian formulation of Wilson's lattice gauge theories, *Phys. Rev. D* **11**, 395 (1975).
- [2] U.-J. Wiese, From quantum link models to D-theory: a resource efficient framework for the quantum simulation and computation of gauge theories, arXiv:2107.09335.
- [3] K. Osterwalder and E. Seiler, Gauge field theories on a lattice, *Ann. Phys.* **110**, 440–471 (1978).